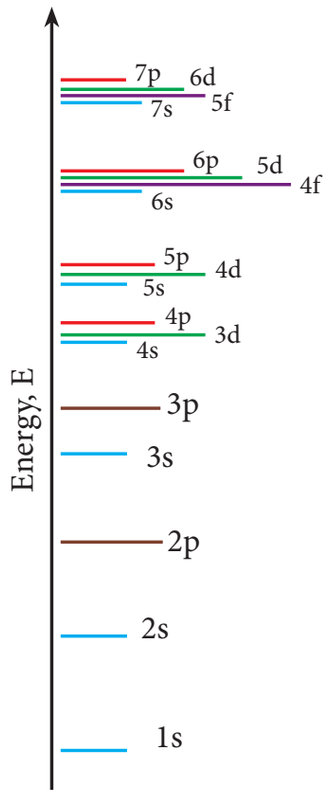


Figure. 2.11 Energy levels of atomic orbitals

The energies of same orbital decrease with an increase in the atomic number. For example, the energy of the 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium and that of lithium is greater than that of sodium and so on, that is, $E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$.

Evaluate Yourself

5. Energy of an electron in hydrogen atom in ground state is -13.6 eV. What is the energy of the electron in the second excited state?

2.6 Filling of orbitals:

In an atom, the electrons are filled in various orbitals according to aufbau principle, Pauli exclusion principle and Hund's rule. These rules are described below.

2.6.1 Aufbau principle:

The word Aufbau in German means 'building up'. In the ground state of the atoms, the orbitals are filled in the order of their increasing energies. That is the electrons first occupy the lowest energy orbital available to them.

Once the lower energy orbitals are completely filled, then the electrons enter the next higher energy orbitals. The order of filling of various orbitals as per the Aufbau principle is given in the figure 2.12 which is in accordance with (n+*l*) rule.

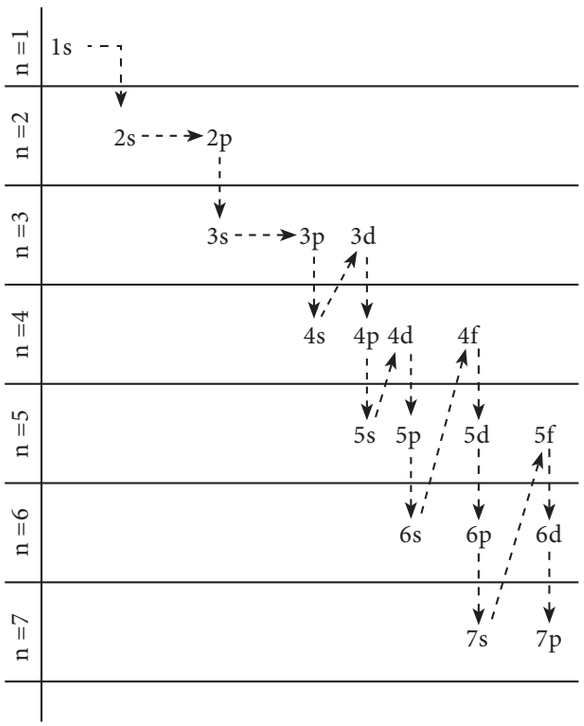


Figure. 2.12 Aufbau principle

2.6.2 Pauli Exclusion Principle :

Pauli formulated the exclusion principle which states that "No two electrons in an atom can have the same set

11th Std Chemistry 038-067.indd 52 06-01-2021 15:06:51